Experiment No. 2

Perform Website, Email, Whois, DNS, and Network Footprinting on the Target Network

AIM

To perform detailed footprinting of a target network by collecting website, email, WHOIS, DNS, and network-level information using various footprinting tools and online utilities.

PROCEDURE

1. Website Footprinting: Use tools such as Burp Suite, HTTrack, or browser developer tools to examine the target website's HTML source, headers, cookies, and directory structure.
2. Email Footprinting: Use email tracking tools (e.g., eMailTrackerPro) to analyze email headers of a received mail and extract the sender's IP address, mail server, and routing path.
3. WHOIS Footprinting: Use WHOIS lookup tools (whois.domaintools.com or the whois command) to gather domain registration details such as registrant name, registrar, creation/expiry dates, and name servers.
4. DNS Footprinting: Use nslookup, dig, or online DNS lookup tools to query A, MX, NS, TXT, and CNAME records of the target domain.
5. Network Footprinting: Use tools such as traceroute/tracert and Path Analyzer Pro to map the network path and identify intermediate hops/routers to the target.
6. Document IP address ranges, ASN information, and network topology revealed from the above steps.
7. Consolidate all findings into a network footprinting report.

SAMPLE OUTPUT

```
$ whois targetcompany.com
Domain Name: TARGETCOMPANY.COM
Registrar: GoDaddy.com, LLC
Creation Date: 2014-03-11
Registry Expiry Date: 2027-03-11
Name Server: NS1.TARGETCOMPANY.COM
Name Server: NS2.TARGETCOMPANY.COM

$ nslookup -type=MX targetcompany.com
targetcompany.com    MX preference = 10, mail exchanger = mail.targetcompany.com

$ tracert targetcompany.com
1    2 ms   1 ms   1 ms  192.168.1.1
2    8 ms   7 ms   9 ms  10.10.0.1
3   15 ms  14 ms  16 ms  203.0.113.1
4   22 ms  21 ms  23 ms  target-edge-router.targetcompany.com
5   25 ms  24 ms  26 ms  203.0.113.45  [target server]
```

RESULT

Website, email, WHOIS, DNS, and network footprinting of the target domain was successfully performed, and key information such as domain registration data, mail server details, DNS records, and the network path to the target was documented.